ic_autolab round 22 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T07:50:02Z at git `15ae5647cbdfb4df0e2a90516f83c33495a2dd50`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-09fa2c` (`python3 tools/pcc/pcc.py standing PCC-20261104-09fa2c`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
32 **improvement**	wsum2-b120-n2	0.838 [0.749, 0.937]	0.837 [0.741, 0.946]	0.833 [0.702, 0.988]	sel < inc	2.519 / 2.759 / 4.915	3.006 / 3.295 / 5.901	0.918 / 1.008 / 1.818	2.7x / 2.7x / 2.7x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

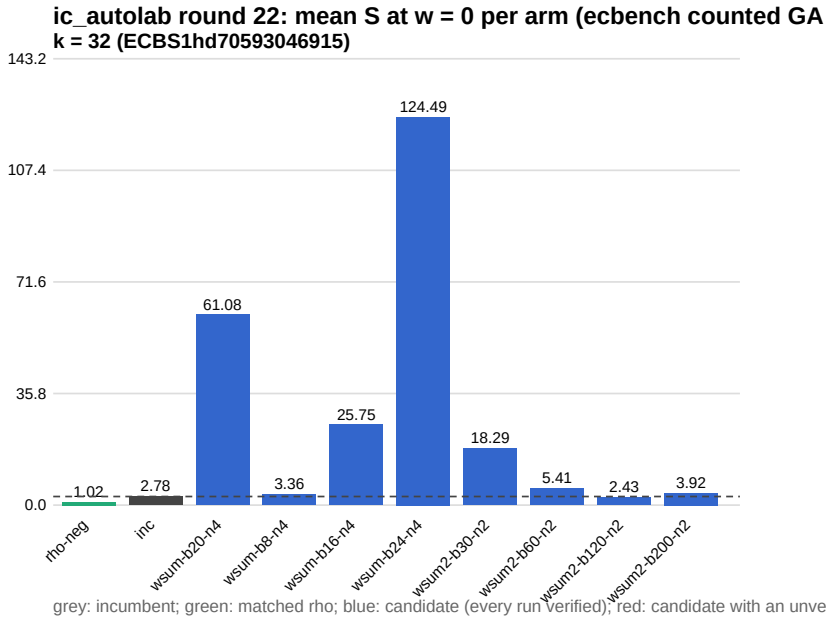
- k = 32: compare ops.status = ok, admissible = true; sel / inc = 0.8381 [0.749, 0.937] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 32: session `ECBS1hd70593046915` (run exit 0, verify exit 0, receipt sha256 `7d7511a185bda23e80c45eeaef1de6f07923e4a49db23c30bfdf5e71beeaadd9`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.018 / 1.118 / 2.018	matched reference
`inc`	wsum-b20	3 `prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=20,free=1`	32	32	2.775 / 3.019 / 5.213	incumbent
`wsum-b20-n4`	wsum-b20	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=20,free=1`	32	32	61.084 / 61.092 / 61.164	incumbent neighbourhood: N = 1.2 x 3
`wsum-b8-n4`	wsum-b8	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=8,free=1`	32	32	3.363 / 3.661 / 6.343	cross-size transfer: wsum-b8-n3 is the k = 24 incumbent at N / N_a = 0.037
`wsum-b16-n4`	wsum-b16	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=16,free=1`	32	32	25.751 / 25.770 / 25.937	alternative wsum-b16 (best S(0) 3.081 at N = 3): N = 1.2 x 3
`wsum-b24-n4`	wsum-b24	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=24,free=1`	32	32	124.487 / 124.492 / 124.532	alternative wsum-b24 (best S(0) 3.369 at N = 3): N = 1.2 x 3
`wsum2-b30-n2`	wsum2-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	18.286 / 21.902 /	exploration: family wsum2-

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
						54.447	b30 untried at k = 32, N = 0.0001 x N_a
wsum2-b60-n2	wsum2-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	5.414 / 6.414 / 15.418	exploration: family wsum2-b60 untried at k = 32, N = 0.0001 x N_a
wsum2-b120-n2	wsum2-b120	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=120,free=1	32	32	2.426 / 2.647 / 4.637	exploration: family wsum2-b120 untried at k = 32, N = 0.0001 x N_a
wsum2-b200-n2	wsum2-b200	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=200,free=1	32	32	3.917 / 3.994 / 4.682	exploration: family wsum2-b200 untried at k = 32, N = 0.0001 x N_a



train screen

Selection

- k = 32: challenger `wsum2-b120-n2` (Pareto alternative at w = 1: `wsum2-b120-n2`)

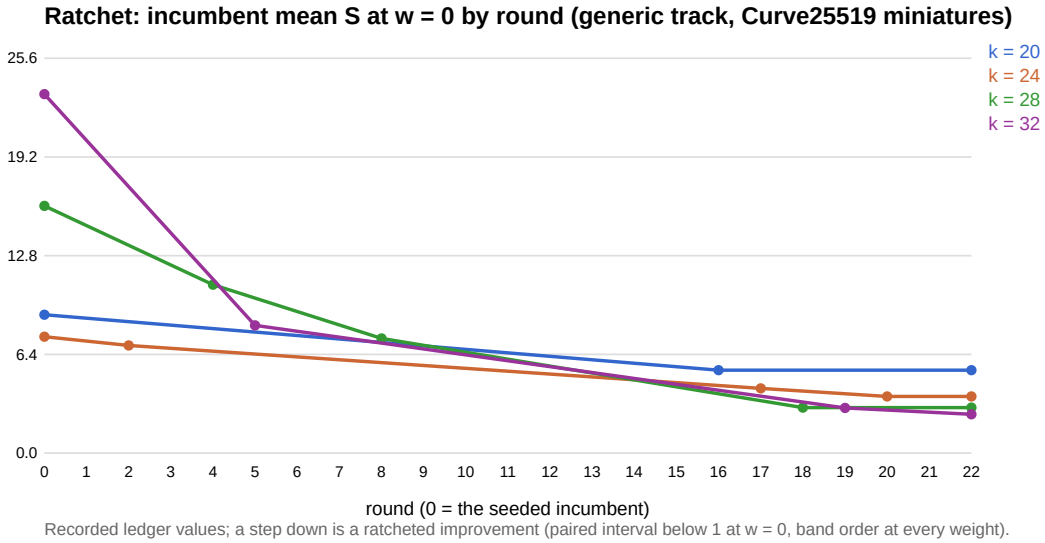
Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
32	ECBS1h66e91aeb7780	0	0	640d6498ac9ac840568cd7523a51a73e3700449933430b4ace96f67d8487bbe3	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	wsum-b5-n3 (prime-projected-	16	5.384 / 5.609 / 7.638	3.6x / 3.5x / 3.0x

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
24	`abscissa:size=3,scan_cap=48 + wsum:coef=5,free=1)` `wsum-b8-n3` (`prime-projected-abscissa:size=3,scan_cap=48 + wsum:coef=8,free=1)`	20	3.678 / 3.920 / 6.100	3.0x / 3.0x / 2.7x
28	`wsum-b12-n3` (`prime-projected-abscissa:size=3,scan_cap=48 + wsum:coef=12,free=1)`	18	2.949 / 3.222 / 5.676	3.0x / 3.0x / 3.0x
32	`wsum2-b120-n2` (`prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=120,free=1)`	22	2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x



ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.